Fiscal Year:	FY 2014	**Task Last Updated:**	FY 06/05/2014
PI Name:	Hsieh, Adam Ph.D.		
Project Title:	Effects of Stem Cell Morphology and Motility on Osteogenesis in Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	20742-5100	**Congressional District:**	5
Comments:			
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	08/01/2013	**End Date:**	07/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:			
Flight Assignment:	NOTE: Extended to 7/31/2015 per A. Chu/ARC and NSSC information (Ed., 8/5/14) NOTE: To be extended, per PI (Ed., 5/28/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Yew, Alvin (NASA Goddard Space Flight Center)		
Grant/Contract No.:	NNX13AM06G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Mesenchymal stem cells (MSCs) are responsible for tissue development, help maintain homeostasis in mature tissues, and are being considered for therapeutic treatments of a variety of cellular lineages including bone, skeletal muscles, and fatty tissues. Chemical and mechanical factors modulate lineage commitment; however, spaceflight can drastically change these cues and hence, influence physiologic processes, such as bone formation. We hypothesize that such changes arise from altered cell-cell interactions and chemotactic behaviors brought about by cell morphologies and cytoarchitectures adopted during microgravity. We seek to investigate the effects of microgravity on MSC morphology, migration and osteogenic differentiation using a newly developed clinorotation time-lapse microscopy (CTM) system and lab-on-chip devices to generate chemical gradients. While the underlying principles of CTM are identical to conventional clinostats, CTM is amenable to live-cell imaging, without prohibitively large experimental setups or disruption of culture environments. Importantly, CTM represents a significant step forward in space biology research, because it enables ground-based microgravity studies of not only traditional end-point outcomes, but also dynamic cellular processes. Specifically, we aim to: (1) investigate the long-term, dynamic behaviors of MSCs under chemokine gradients in simulated microgravity by analyzing the time-dependent alterations to morphology, motility, and cell density and (2) quantify changes in distributions of focal adhesions and osteogenic phenotype indicators by using CTM to monitor a live-cell alkaline phosphatase (AP) stain and end-point vinculin, lipids, and AP stains. This investigation provides novel insight into how dynamic MSC events are affected by microgravity. We believe that our hypothesis may partially explain a mechanism responsible for reduced bone mass and impaired fracture healing in space. Spinoff research includes: investigating genomic and proteomic profiles associated with MSC cytoarchitecture and differentiation; assessing the potential of MSCs in treating musculoskeletal pathologies in spaceflight; using drugs to promote normal MSC-mediated morphogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Spinoff research includes: investigating genomic and proteomic profiles associated with MSC cytoarchitecture and differentiation; assessing the potential of MSCs in treating musculoskeletal pathologies in spaceflight; using drugs to promote normal MSC-mediated morphogenesis.
Task Progress:	Overall, we have made some important strides toward the completion of the proposed project. Our progress has been productive but limited, due to the marginal delay in award selections that challenged our efforts to secure research personnel within the standard graduate student recruiting cycle in the Department of Bioengineering. However, we have notably demonstrated a threshold response of human mesenchymal stem cell (hMSC) morphology on clinorotation speed, one key aspect in Aim 1 of the project. We also found that focal adhesion formation in hMSCs may be distinct between static and simulated microgravity in pilot experiments pertaining to Aim 2. To remedy our personnel issues, we recently hired a postdoctoral fellow with high interest and research compatibility to spearhead experimental efforts full-time, which we believe will greatly improve our progress. A one year extension was requested, and we anticipate being able to complete the project objectives within this timeframe.
Bibliography Type:	Description: (Last Updated: 10/30/2015)
Abstracts for Journals and Proceedings	Yew AG, Hsieh AH. "Frequency response of stem cells to gravity loads." 7th World Congress of Biomechanics, Boston, MA, July 6-11, 2014. 7th World Congress of Biomechanics, Boston, MA, July 6-11, 2014. , Jul-2014
Articles in Peer-reviewed Journals	Yew AG, Atencia J, Hsieh AH. "Lab-on-chip clinorotation system for live-cell microscopy under simulated microgravity." Cellular and Molecular Bioengineering. 2014 Mar;7(1):165-70. http://dx.doi.org/10.1007/s12195-013-0319-2 , Mar-2014